

Module - 2.

Exploratory data Analysis (EDA)

Q.1. Descriptive Statistics: Mean, median, mode, Variance, Standard deviation, percentiles using pandas and Scipy.

Exploratory Data Analysis (EDA) is the process of Examining, Summarizing and understanding a dataset before applying Statistical analysis.

EDA mainly use descriptive Statistics and data visualization.

Descriptive Statistics:

Descriptive Statistics are methods used to Summarize and describe the Important characteristics of a dataset.

The major descriptive Statistical measures are.

1. Mean
2. Median
3. Mode
4. Variance
5. Standard Deviation
6. Percentiles.

Two Important library.

1. Pandas: Pandas is a python library used for data handling, manipulation and analysis.

Import pandas as pd

② Scipy

Scipy stands for Scientific python. It is a python library used for scientific and statistical calculations.

from Scipy import stats.

Scipy provides statistical functions through scipy.stats.

Pandas

mean()
median()
mode()
var()
std()
quantile()
describe()

Scipy

stats.^{np.}mean()
stats.^{np.}median()
stats.mode()
np.var()
np.std()
np.percentile()
stats.describe()

Code: using pandas.

```
import pandas as pd
# create the dataset
marks = pd.Series([45, 50, 35, 60, 60, 65, 70, 75, 80, 90])
# Descriptive Statistics
print("Mean:", marks.mean())
print("Median:", marks.median())
print("Mode:", marks.mode())
print("Variance:", marks.var())
print("Standard Deviation:", marks.std())
# Minimum and Maximum
print("Minimum:", marks.min())
print("Maximum:", marks.max())
```

```
# Percentile  
print("25th percentile:", marks.quantile(0.25))  
print("50th percentile:", marks.quantile(0.50))  
print("75th percentile:", marks.quantile(0.75))
```

```
print(marks.describe()) # complete Summary
```

Code using Scipy

```
import numpy as np  
from scipy import stats
```

```
marks = np.array([ ])
```

```
print("Mean:", np.mean(marks))  
print("Median:", np.median(marks))  
print("Mode:", np.mode(marks))  
print("Variance:", np.var(marks))  
print("Standard Deviation:", np.std(marks))
```

```
print("Minimum:", np.min(marks))  
print("Maximum:", np.max(marks))
```

```
print("25th percentile:", np.percentile(marks, 25))  
print("50th percentile:", np.percentile(marks, 50))  
print("75th percentile:", np.percentile(marks, 75))  
print(stats.describe(marks))
```

Note: Pandas use Quantile(), while Numpy use percentile(),
because pandas use a decimal value like 0.25, while
Numpy use a percentage value like 25.

1. Mean

The mean is the arithmetic average of all values in a dataset.

Formula:

$$\text{Mean} = \frac{\sum x}{n}$$

where,

x = Individual observation

n = number of observation.

Ex: 10, 20, 30, 40, 50

$$\text{Mean} = \frac{10 + 20 + 30 + 40 + 50}{5} = \frac{150}{5} = 30.$$

a. Median

The median is the middle value when the data is arranged in ascending or descending order.

Case 1: Odd number of observations.

10, 20, 30, 40, 50.

The middle value = 30

$\therefore$ median = 30.

Case 2: Even number of observations

10, 20, 30, 40.

There are two middle values: 20 & 30.

$$\text{median} = \frac{20 + 30}{2} = 25.$$

$\therefore$, median = 25.

3. Mode.

The mode is the value that occurs most frequently in a dataset.

Mode can be used for both numerical and categorical data.

Ex: Data = 10, 20, 20, 30, 40, 20, 50.

Here, 20 occurs three times,

therefore mode = 20.

4. Percentile.

percentile is the position of a value compared with all the other values in the dataset.

Important percentiles

25th percentile.

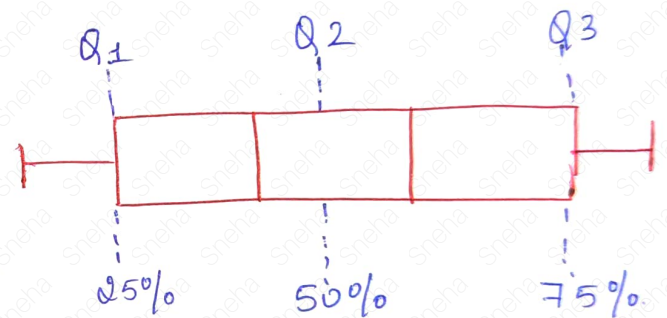
Q_1

50th percentile.

median / Q_2

75th percentile.

Q_3



here, how do we actually calculate 25th, 50th, 75th percentile

Take the data: 45, 50, 55, 60, 60, 65, 70, 75, 80, 90

Arrange the data:

Position	1	2	3	4	5	6	7	8	9	10
Value	45	50	55	60	60	65	70	75	80	90

Calculate the 25th percentile.

$$\text{position} = \frac{P}{100}(n-1) + 1$$

where P = percentile.

n = number of observations.

Here $P = 25$, $n = 10$

$$\begin{aligned}\therefore \text{Position} &= \frac{25}{100}(10-1) + 1 \\ &= 0.25(9) + 1 \\ &= 2.25 + 1 \\ &= \underline{3.25}\end{aligned}$$

So, the 25th percentile lies between the 3rd and 4th values.

3rd value = 55

4th value = 60

Now take 25% of the difference.

$$60 - 55 = 5$$

$$25\% \times 5 = \underline{1.25}$$

Add the 1.25 value to the 55 because it is the starting point of the 3rd value.

$$55 + 1.25 = 56.25$$

Therefore

$$P_{25} = \underline{56.25}$$

Calculate the 50th percentile.

The 50th percentile is the median.

So, the middle value is

$$5^{\text{th}} \text{ value} = 60$$

$$6^{\text{th}} \text{ value} = 65$$

$$P_{50} = \frac{60+65}{2} = \frac{125}{2}$$

$$P_{50} = \underline{\underline{62.5}}$$

Calculate the 75th percentile.

$$\text{Position} = \frac{P}{100} (n-1) + 1$$

where P = percentile.

n = number of observation

$$\text{Here } P = 75, n = 10$$

$$\begin{aligned}\therefore \text{Position} &= \frac{75}{100} (10-1) + 1 \\ &= 0.75 (9) + 1 \\ &= 6.75 + 1 \\ &= 7.75\end{aligned}$$

So, the 75th percentile lies b/w 7th & 8th value.

$$7^{\text{th}} \text{ value} = 70$$

$$8^{\text{th}} \text{ value} = 75$$

$$75 - 70 = 5$$

$$75\% \times 5 = 3.75$$

$$70 + 3.75 = 73.75$$

Therefore,

$$P_{75} = \underline{\underline{73.75}}$$

Variance

Variance is a statistical measure that tells how much the values in a dataset are spread out from their mean (average).

low variance $\rightarrow$ values are close to the mean

High Variance $\rightarrow$ values are far from the mean.

For Example: Consider two sets of marks.

Set A: 48, 49, 50, 51, 52

Set B: 20, 35, 50, 65, 80

Both have a mean of 50, but Set B is much more spread out.

Calculation of the Variance.

Ex: Take 2, 4, 6.

Step 1: Find the mean

$$\text{mean} = \frac{2+4+6}{3} = 4$$

Step 2: Find the difference from mean.

Value	Difference from mean
2	$2 - 4 = -2$
4	$4 - 4 = 0$
6	$6 - 4 = +2$

Step 3: Square the difference.

Value	difference	Square difference.
2	-2	4
4	0	0
6	+2	4

Step 4: Add the Squared differences

$$4 + 0 + 4 = 8$$

Step 5: Divide by number of values.

$$\text{Variance} = 8/3$$

$$= \underline{\underline{2.67}}$$

So, the Variance of 2, 4, 6 is 2.67.

Standard Deviation

Standard deviation is the Square root of variance.
Standard deviation tells approximately how far the data values are spread from the mean.

In previous example,

We got Variance = 2.67

$$SD = \sqrt{2.67}$$

$$\approx \underline{\underline{1.63}}$$

2.2. Data distributions: normal, Skewed, Kurtosis; probability distributions

Data distributions:

Data distributions refers to the way in which the values of a dataset are spread, concentrated, (or) arranged across different values (or) ranges.

In data analysis, we usually collect a large amount/ numbers of observations. These observations may not have the same values. Some values may be small, some may be large and many values may be concentrated around a particular range.

For example, Suppose we collect the marks of 20 students:
35, 42, 45, 48, 50, 52, 53, 55, 57, 58, 60, 61, 62, 63, 64, 65, 67, 70, 72, 85.

The marks are not Equally distributed. Some students have marks in the 40's, many students have marks between 50 and 65, and only a few students have very high marks such as 72 and 85.

The pattern in which these observations are spread is called data distribution.

Importance of data distribution.

Data distribution is important because statistical analysis does not depend only on the average value. Two datasets can have the same mean but can have completely different patterns.

Consider two classes.

Class A $\rightarrow$ 55, 57, 58, 60, 61, 62, 65

Class B $\rightarrow$ 20, 30, 40, 60, 80, 90, 100

Class A, most values are close to 60.

Class B, values are widely spread

Both datasets may have a similar average, but their distributions are different.

Therefore, studying the distributions helps us to understand:

1. Where most observations are concentrated.
2. How widely the observations are spread.
3. Whether the data is symmetric (or) asymmetric.
4. Whether the data contains outliers.
5. What statistical methods may be used.

Characteristics of a Distribution.

1. Center

The center tells where the data is generally located.

Common measures are:

1. Mean
2. Median
3. Mode

2. Spread

Spread tells how far the observations are from one another.

Common measures are:

1. Range.
2. Variance.
3. Standard deviation
4. Interquartile range.

3. Shape:

Shape explains the overall pattern of the distribution.

Ex: 1. Symmetric

2. Normal

3. Right-skewed.

4. Left-skewed.

4. Outliers

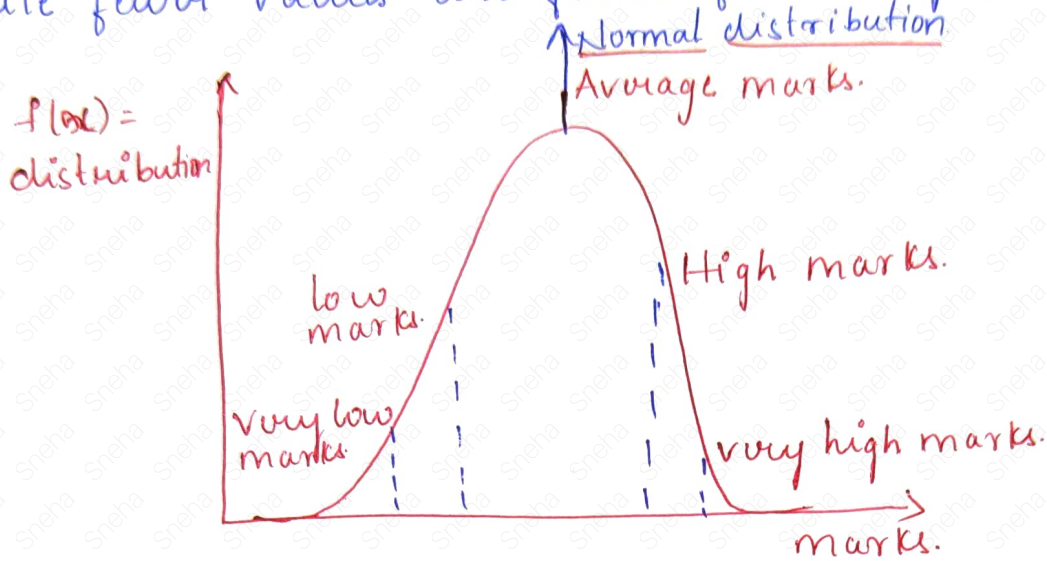
An outlier is an unusually high (or) low value compared with the other observations.

Ex: 45, 48, 50, 52, 53, 55, 95

Here 95 may be an outlier because it is far away from most of the other values.

1. Normal Distribution

Normal Distribution is a Symmetric, bell-shaped distribution in which data values are near the average, while fewer values are found far away from the average.

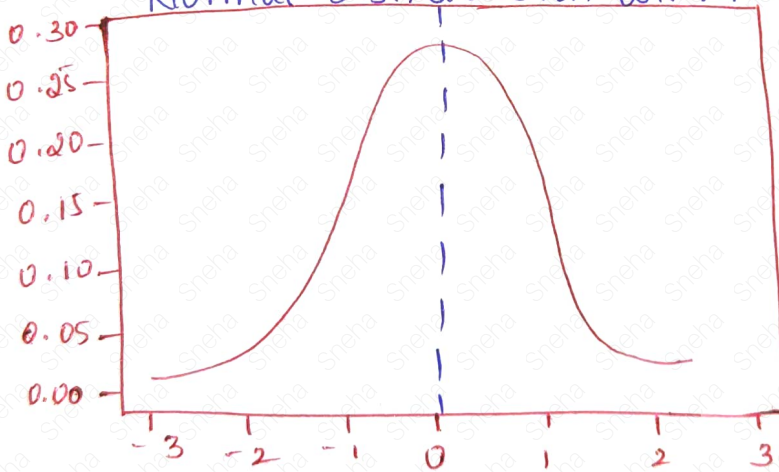


Formula for Normal distribution.

$$\text{Mean} = \text{Median} = \text{Mode.}$$

Normal Distribution with mean, median and mode.

Example:



Example:

Suppose, measure the heights of 1,000 students. will most student have extremely low or extremely high height?

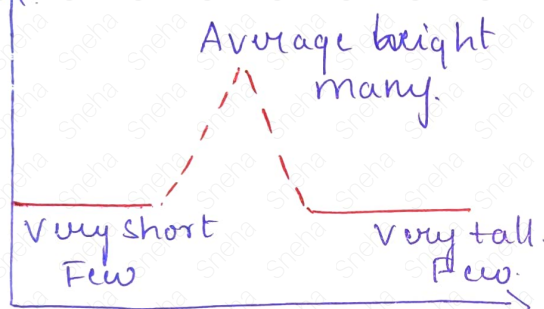
The answer is No, most student will be somewhere around the average.

150 155 160 165 {170} 175 180 185 190

Most student are around 170cm. very few people are at the extreme ends. we can divide them into three groups:

So, the basic pattern is

Number of students



1. Very short $\rightarrow$ Few

2. Average height $\rightarrow$ many

3. Very tall $\rightarrow$ Few.

* The middle of the curve is high because many students have heights close to the average.

* The two ends are low because only a few people are extremely short (or) extremely tall.

Thus, The distribution has a bell-shape.

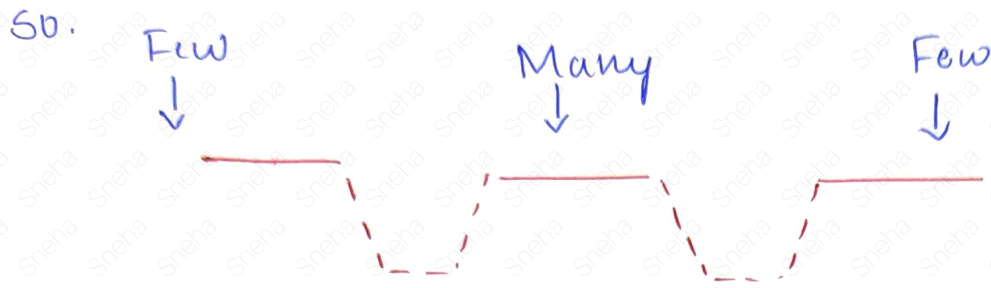
Connecting it to Mean, median and mode.

In an normal distribution, the center of the bell curve represents:

mean = median = mode.

Suppose the average height is 170 cm.

Most people may have height around 170 cm, while fewer people may have height such as 140 cm (or) 200 cm.



Mean = Median = mode.
around 170 cm.

Characteristics of Normal distributions

1. Symmetry: The normal distribution is symmetric around its mean. This means the left side of the distribution mirrors the right side.
2. Mean, Median and Mode: In a normal distribution, the mean, median and mode are all equal and located at the center of the distribution.
3. Bell-shaped curve: The curve is bell-shaped, indicating the most of the observations cluster around the central peak and the probabilities for values further away from the mean equally in both directions.
4. Standard Deviation: Standard deviation explains how far the data values are spread around the average.

2. Skewness

Skewness is a statistical measure that shows how data is spread out in a dataset. It tells if the data points are skewed to the left (negative skew) or to the right (positive skew) in relation to the mean.

It is important because it helps us to understand the shape of the data distribution for accurate data analysis and helps in identifying outliers & findings the best statistical methods to use for analysis.

Types of Skewness

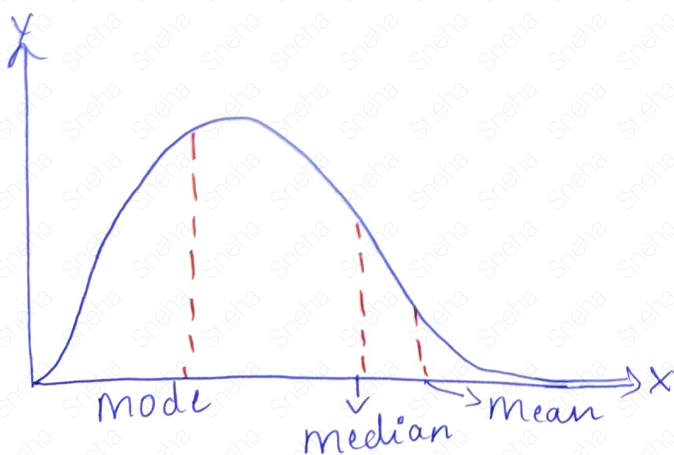
1. positive skewness (Right skew/tail)
2. Negative skewness (Left skew/tail)
3. Zero skewness (Symmetrical Distribution)

1. Positive Skewness

In positive skewness, the right tail is longer than the left which means most data points are on the left with a few large values pulling the distribution to the right.

Relationship

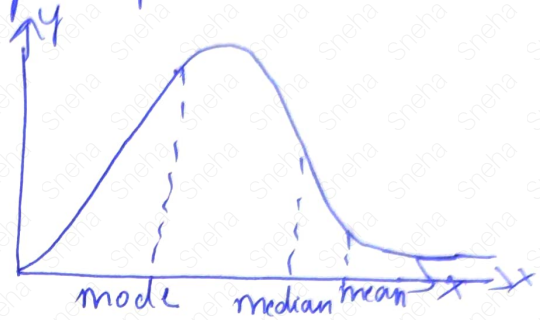
$$\text{Mean} > \text{Median} > \text{Mode}$$



Example: Consider the monthly Income of 10 people.

20K, 22K, 23K, 24K, 25K, 26K, 27K, 28K, 30K, 2Lakh.

Most people earn between 20K and 30K, but one person earns 2Lakh. That very large value creates a long tail on the right.

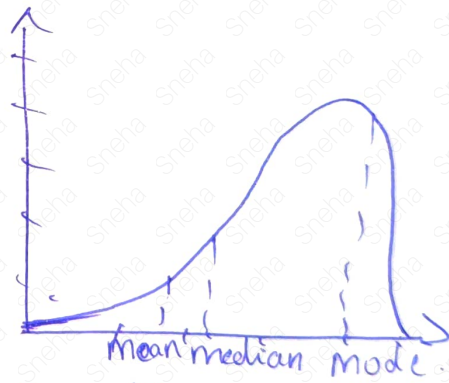


2. Negative Skewness

In Negative Skewness, the left tail is longer which means most data points are on the right with a few smaller values pulling the distribution to the left.

Relationship:

$$\text{Mean} < \text{Median} < \text{Mode}$$



Example: Consider the Students marks

20, 75, 80, 82, 85, 86, 88, 90, 92, 95.

most students scored between 75 and 95 but one student scored only 20. That very low value creates a long tail toward the left.

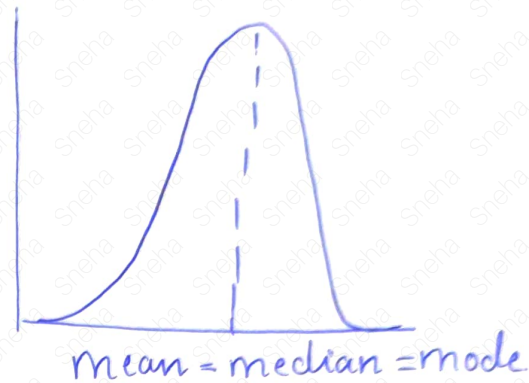
3. Zero Skewness

Zero skewness shows a perfectly symmetrical distribution where the mean, median and mode are equal.

Relationship :

$$\text{Mean} = \text{Median} = \text{Mode}.$$

Example: Suppose the students marks are: 40, 50, 60, 70, 80.



Here, the values are balanced around 60.

$$\text{Mean} = 60.$$

$$\text{Median} = 60.$$

$$\text{mode} = \text{No mode}.$$

The distribution is approximately symmetrical.

Measurement of Skewness

Skewness is measured using different techniques to quantify the degree of asymmetry in a dataset's distribution.

There are three methods to measure skewness.

1. Karl Pearson's Measure.

2. Bowley's Measure.

3. Kelly's Measure.

1. Karl Pearson's Measure.

Karl Pearson's Measure uses the mean, median and standard deviation of the given data to measure the asymmetry of the distribution.

Formula:

1. with respect to Mean and Median.

$$S_K = \frac{3 \times (\bar{X} - M)}{\sigma}$$

2. with respect to Mean and Mode.

$$S_K = \frac{(\bar{X} - \text{Mode})}{\sigma}$$

where,

S_K = Karl Pearson's Skewness coefficient

$\bar{X}$ = Mean of the dataset

M = Median of the dataset

σ = Standard deviation of the dataset.

Interpretation:

when Skewness = 0 ; The distribution is Symmetric.

Skewness > 0 ; The distribution is positive.

Skewness < 0 ; The distribution is negative.

Example:

Calculate Pearson's skewness coefficient for a dataset of Exam Scores: 85, 88, 92, 94, 96, 98, 100, 100, 100, 100

→ Step 1: Calculation of Mean.

$$\text{Mean } (\bar{x}) = \frac{\sum x}{n} = \frac{85+88+\dots}{10} = \frac{953}{10} = 95.3$$

Step 2: Calculation of Median.

$$\text{Median} = \frac{96+98}{2} = \frac{194}{2} = 97$$

Step 3: Calculation of Standard deviation.

$$\sigma^2 = \frac{\sum (x_i - \mu)^2}{N} = \frac{(85-95.3)^2 + \dots + (100-95.3)^2}{10}$$

$$= \frac{268.1}{10} = 26.81$$

$$\sigma^2 = 26.81$$

$$\sigma = \sqrt{26.81}$$

$$\sigma = 5$$

Step 4: Calculation of mode.

100 is the most frequently occurring value in the data.

Step 5: Substitute the values in the formulae.

1. with respect to mean & median.

$$S_k = \frac{3(\bar{x} - M)}{\sigma} = \frac{(3 \times (95.3 - 97))}{5} = \frac{-5.1}{5}$$

$$S_k = -1.02$$

2. with respect to mean and mode.

$$S_k = \frac{(\bar{x} - \text{mode})}{\sigma} = \frac{(95.3 - 100)}{5}$$

$$S_k = -0.94$$

Since the skewness coefficient (S_k) is negative which shows a slight negative skewness in the distribution of Exam scores.

2. Bowley's Measure.

Bowley's skewness is another method for calculating skewness based on quartiles (Q_1, Q_2, Q_3).

Formula:

$$B = \frac{Q_3 + Q_1 - 2Q_2}{Q_3 - Q_1}$$

where $Q_1 \rightarrow$ First Quartile (25th percentile)

$Q_2 \rightarrow$ Second Quartile.

$Q_3 \rightarrow$ Third Quartile.

Interpretation:

$B = 0$: Symmetric

$B > 0$: positive.

$B < 0$: negative.

Ex: Calculate Bowley's Measure of Skewness for the following dataset representing the ages of a group of people.
20, 24, 28, 32, 35, 40, 42, 45, 50.

→ Step 1: Calculate the median (Q_2)

$$Q_2 = 35$$

Step 2: Calculate the first Quartile (Q_1)

To find Q_1 , let's consider the values to the left: 20, 24, 28, 32.

$$Q_1 = \frac{24+28}{2} = 26.$$

Step 3: Calculate the third Quartile (Q_3).

$$Q_3 = \frac{42+45}{2} = 43.5.$$

Step 4: Substitute the above values in the formula.

$$\begin{aligned} B &= \frac{Q_1 + Q_3 - 2Q_2}{Q_3 - Q_1} \\ &= \frac{26 + 43.5 - 2 \times 35}{43.5 - 26} \\ &= \underline{\underline{-0.02}} \end{aligned}$$

Since $B < 0$, this shows a negatively skewed.

3. Kelly's Measure.

Kelly's skewness measure calculates skewness by comparing certain percentiles in the data which typically the 10th, 50th (median) and 90th percentiles.

Formula:

$$\text{Skewness} = \frac{P_{90} + P_{10} - 2P_{50}}{P_{90} - P_{10}}$$

Ex: Calculate Kelly's coefficient of skewness for the following data: 5, 7, 8, 9, 10, 12, 15, 16, 18, 20.

→ Step 1: Find the 10th percentile.

Arrange the data ascending order, so the percentile is 5.

$$P_{10} = 5$$

Step 2: Find the 50th percentile (median).

$$= \frac{10 + 12}{2} = 22/2 = 11$$

$$P_{50} = 11$$

Step 3: Find 90th percentile.

$$P_{90} = 18$$

Step 4: Substitute in the formula.

$$\begin{aligned} & \frac{P_{90} + P_{10} - 2P_{50}}{P_{90} - P_{10}} \\ &= \frac{18 + 5 - 2(11)}{18 - 5} = 0.07 \end{aligned}$$

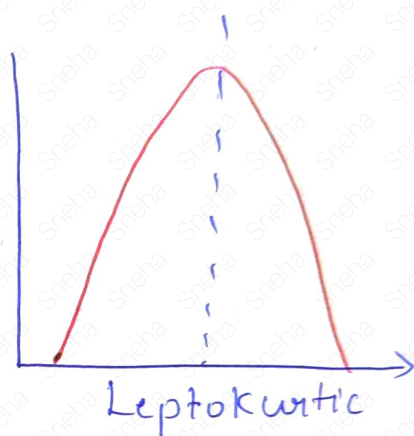
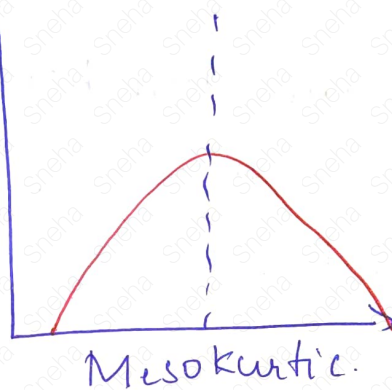
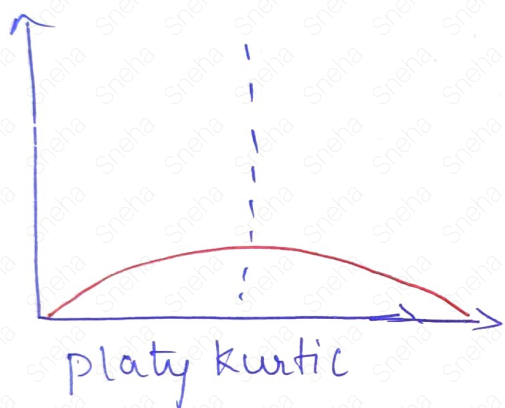
Since $SKL > 0$, this shows a positive skewness.

3. Kurtosis

Kurtosis is a statistical measure that describes the shape of a data distribution especially how heavy or light the tails are.

Type of Kurtosis

1. Leptokurtic
2. Mesokurtic
3. platykurtic



Example of Kurtosis

Suppose we have a two datasets

A : 45, 48, 50, 52, 54, 55, 57.

B : 20, 48, 50, 52, 54, 55, 90.

From A there are no very extreme values.
while B have a extreme values.

20
↓
Extreme

90.
↓
Extreme.

Kurtosis helps us describe the tail behaviour & extreme observations.

1. Mesokurtic

A distribution with tail behavior approximately like a normal distribution is called mesokurtic.

2. Leptokurtic

A leptokurtic distribution has heavier tails, meaning extreme observations are more prominent.

leptokurtic = Heavy tails.

3. platykurtic

A platykurtic distribution has lighter tails, meaning extreme observations are less prominent.

platykurtic = light tails.

Probability Distribution

A probability distribution is a statistical function that shows how the possible values of a random variable are distributed and the probability associated with each value.

Random variable is a variable whose value depends on the outcome of a random experiment.

Example :

1. probability distribution of a coin.

A coin is tossed once.

possible outcomes = Head (H) or Tail (T).

probability distribution table.

Outcome (x)

probability $P(x)$

Head (H)

$$1/2 = 0.5$$

Tail (T)

$$1/2 = 0.5$$

$$\text{Total probability} = 0.5 + 0.5 = 1.$$

2. probability distribution of Tossing Two coins.

possible outcomes = HH, HT, TH, TT.

probability distribution table.

No of Heads (x)

outcomes

probability $P(x)$

0

TT

$$1/4 = 0.25$$

1

HT, TH

$$2/4 = 0.50$$

2

HH

$$1/4 = 0.25$$

3. probability distribution of a Dice.

A fair die is rolled once.

possible outcomes = 1, 2, 3, 4, 5, 6

probability distribution table.

Outcome (x)

probability $P(x)$

1

$$1/6$$

2

$$1/6$$

3

$$1/6$$

4

$$1/6$$

5

$$1/6$$

6

$$1/6$$

* A die has 6 equally likely outcomes.

* probability of getting is $1/6$

* Total probability = 1.

2.3. Correlation and Covariance; hypothesis testing basis (t-test, chi-square)

Correlation

* Correlation is a statistical measure that describes the direction and strength of the relationship between two numerical values.

* Correlation tells whether two variables move together, in which direction and how strongly they are linearly related.

For Example: Study Hours $\leftrightarrow$ Marks.

Correlation Value

Correlation is usually represented by ' r '

Its value lies between:

$$-1 \leq r \leq 1$$

So correlation can be between -1 and $+1$.

① If $r = +1$, perfect positive correlation.

When one variable increases, the other increases in a perfectly linear way.

Example: Study hours $\uparrow \rightarrow$ Marks $\uparrow$

② If $r = -1$, perfect negative correlation

When one variable increases, the other decreases in a perfectly linear way.

Example: Speed $\uparrow \rightarrow$ Travel time $\downarrow$

③ ~~No or weak~~ $r \approx 0$, no linear correlation.

There is no clear relationship between the variables.

Types of Correlation:

1. positive correlation
2. Negative correlation
3. No or weak correlation.

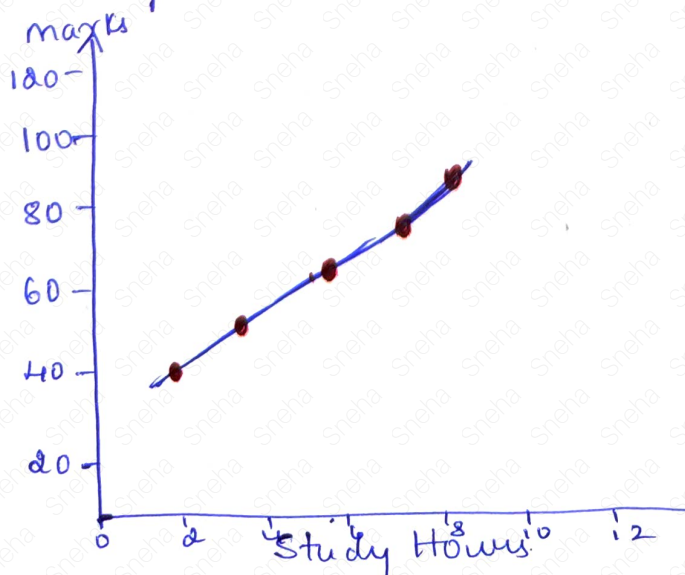
1. positive correlation

When one variable increases, the other variable also tends to increase.

Ex: Study Hours and Marks.

Study Hours	Marks.
2	40
4	50
6	60
8	72
10	85

When study hours increase, the marks also increase, so, it is a positive correlation.



2. Negative Correlation

Negative correlation occurs when one variable increases while the other variable tends to decrease.

Ex: Speed and Travel time.

Speed (km/h)

Travel time (hours)

20

5.0

30

3.33

40

2.50

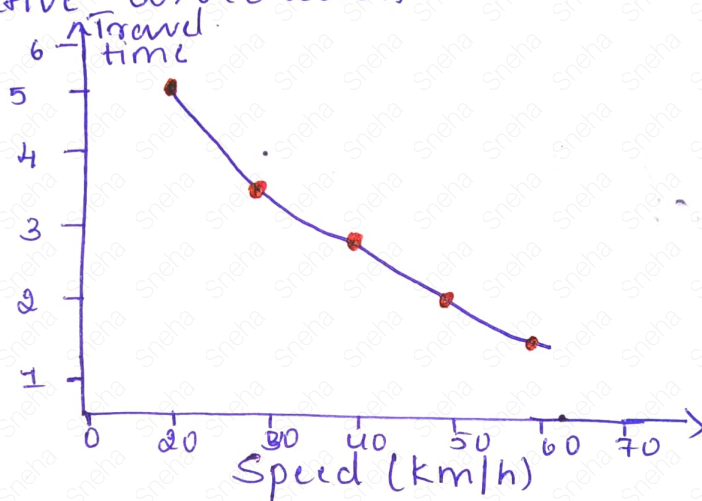
50

2.00

60

1.67

When Speed increases, the travel time decreases, so, it is a negative correlation.



3. No or weak Correlation

When there is no clear relationship between two variables, they have approximately;

$$r=0$$

Ex: Shoe size $\leftrightarrow$ Exam marks.

A student with a shoe size does not necessarily have higher (or) lower marks.

Student

Shoe Size

Exam marks.

A

6

72

B

7

55

C

8

88

D

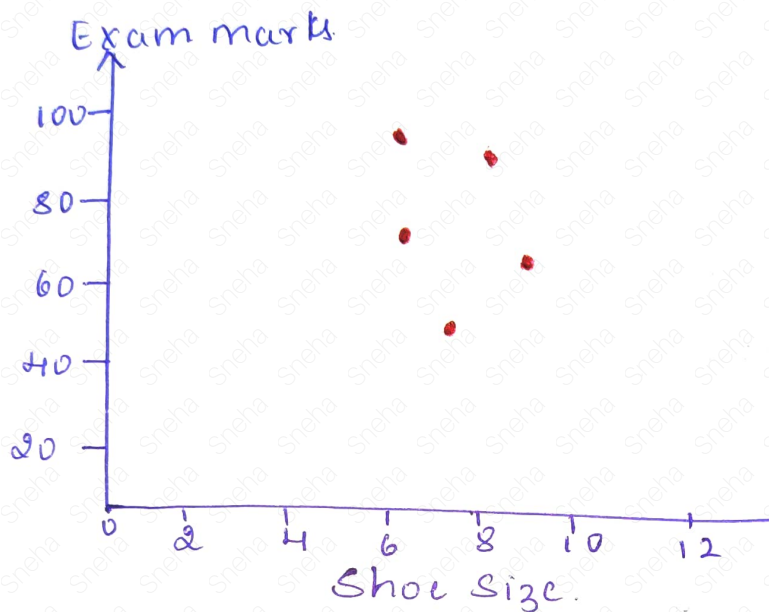
9

61

E

6

94



Covariance

Covariance is a statistical measure that indicates the direction in which two variables change together.

Types of Covariance

1. positive covariance.
2. Negative covariance.
3. Zero covariance.

1. positive Covariance

positive covariance occurs when two variables tend to move in the same direction.

X Increases $\rightarrow Y$ tends to Increase.

X decreases $\rightarrow Y$ tends to decrease.

Example: Rainfall and umbrella sales.

Rainfall

2

5

8

12

15

Umbrella sold

10

20

35

50

70

Both are Increasing therefore it is a positive.

2. Negative Covariance

Negative covariance occurs when two variables tend to move in opposite directions.

X Increases $\rightarrow Y$ tends to decrease.

X decreases $\rightarrow Y$ tends to Increase.

Example: Temperature and Heater usage.

Temp($^{\circ}$ C)

10

15

20

25

30

Heater usage (hours)

8

6

4

2

1

one Increases while the other decreases therefore it is a negative.

3. Zero Covariance

When there is no clear linear pattern between two variables.

One variable may increase or decrease, but the other variable does not show a consistent pattern.

Example: Number of pages and daily Temperature.

Pages	Temperature (°C)
100	25
200	18
300	30
400	22
500	27

There is no consistent, therefore.

Covariance ≈ 0 .

Code:

```
import pandas as pd
# Create data
data = {
    "Study_Hours": [2, 4, 6, 8, 10],
    "Marks": [40, 50, 60, 72, 85]
}
df = pd.DataFrame(data)
print(df)
# Calculate Covariance
covariance = df["Study_Hours"].cov(df["Marks"])
# Calculate Correlation
correlation = df["Study_Hours"].corr(df["Marks"])
print("Covariance:", covariance)
print("Correlation:", correlation)
```

Hypothesis Testing.

* Hypothesis Testing is a Statistical method used to make a decision about a population using Information collected from a Sample.

* Hypothesis testing helps to decide whether a Statement about data is Supported by Evidence (or) not.

Example: Suppose a college says,

"The average mark of Student is 60".

Here, we cannot check Every Student in a large population. Instead, we select a Sample of Students & use their marks to test the Statement.

The process is called hypothesis testing.

There are mainly two hypothesis.

1. Null Hypothesis (H_0)

2. Alternative Hypothesis (H_1)

1. Null Hypothesis (H_0).

The Null hypothesis, is a Statement that there is no Significant, relationship (or) Effect.

2. Alternative Hypothesis (H_1)

The Alternative Hypothesis is a Statement that there is a Significant^{difference}, relationship (or) Effect.

Example:

Suppose a teacher introduces a new teaching method.

Here,

the teacher wants to know whether the new method improves student's marks.

In H_0 ,

The new teaching method does not improve student's marks.

and

H_1 ,

The new teaching method improves student's marks.

$H_0 \rightarrow$ No Improvement.

$H_1 \rightarrow$ Improvement.

Important terms

1. P-value

The p-value decides whether the result is statistically significant.

Decision rule

Condition

$$p \leq 0.05$$

$$p > 0.05$$

Decision

Reject H_0

Fail to Reject H_0

Example:

$$p\text{-value} = 0.02$$

Since

$$0.02 < 0.05$$

Reject H_0 .

$$p\text{-value} = 0.12$$

Since

$$0.12 > 0.05$$

Fail to reject H_0

2. Significance level.

The Significance level is represented by α (alpha)

The commonly used value is:

$$\alpha = 0.05$$

This is called the 5% Significance level.

Ex:

$$\rightarrow \alpha = 0.05$$

and the calculated p-value is

$$p = 0.03$$

Since

$$0.03 < 0.05$$

result: Reject H_0 .

There are two Important tests:

1. t - test.

2. Chi-Square-test.

* t - test mainly used with numerical data and mean.

* Chi-Square used with categorical data.

Ex: T-Tut.

Suppose the marks of 5 Students are:

52, 48, 55, 50, 45.

The teacher wants to know whether the average marks of Student is 60.

Step 1: State the hypothesis.

Null hypothesis (H_0)

The mean is 60.

$$H_0: \mu = 60.$$

Alternative hypothesis (H_1)

The mean is not 60.

$$H_1: \mu \neq 60.$$

Step 2: Calculate the Sample mean.

$$\bar{x} = \frac{\sum x}{n}$$

$$= \frac{55 + 60 + 65 + 70 + 50}{5} = \frac{300}{5}$$

$$= 60$$

Step 3: Calculate the deviations.

mark (x)

x - mean

(x - mean)²

55

-5

25

60

0

0

65

5

25

70

10

100

50

-10

100

250.

Step 4: Calculate Sample Variance.

Formula:
$$S^2 = \frac{\sum (x - \bar{x})^2}{n-1}$$

$$S^2 = \frac{250}{5-1} = \frac{250}{4}$$

$$S^2 = 62.5$$

Step 5: Calculate Standard deviation.

$$S = \sqrt{62.5}$$

$$S = 7.906$$

Step 6: Calculate the t-value.

$$t = \frac{\bar{x} - \mu_0}{S/\sqrt{n}}$$

Where

$\bar{x}$ = Sample mean.

μ_0 = hypothesized mean.

S = standard deviation

n = Sample Size.

$$t = \frac{60 - 60}{7.91\sqrt{5}}$$

$$t = \frac{0}{3.54}$$

$$\boxed{t = 0}$$

Step 7: Calculate degrees of freedom & critical T-value.

$$df = n - 1$$

$$= 5 - 1$$

$$= 4$$

$$\boxed{df = 4}$$

T-table

df	0.05
1	12.706
2	4.303
3	3.182
4	2.776
5	2.571

Step 8: Find the critical T-value.

we have: $df = 4$.

and $\alpha = 0.05$

So, $t_{\text{critical}} = 2.776$.

Step 9: Compare t with critical t .

$$|t| < 2.776$$

So, the result

Step 10: Find p-value.

$$t = 0 \quad df = 4$$

$$p = 1.000$$

Step 11: Decision

compare p-value with significance level.

$$1.000 > 0.05$$

Therefore Do not reject H_0 .

There is no statistically significant evidence that the average marks are different from 50.

② Chi-Square Test

The chi-Square Test is a statistical hypothesis test used to analyze categorical data.

It compares the observed frequencies with the expected frequencies to determine whether there is a statistically significant difference (or) relationship.

The chi-Square test is commonly used in data analytics to study whether two categorical variables are independent (or) related.

Hypothesis

1. Null Hypothesis (H_0)

H_0 : The two categorical variables are independent.

This means there is no significant relationship b/w the variables.

2. Alternate Hypothesis (H_1)

H_1 : The two categorical variables are related.

there is a significant relationship b/w the variables.

Types of Chi-Square Tests

1. Chi-Square Test of Independence

It is used to determine whether two categorical variables are related.

Ex: Attendance and Examination result.

2. Chi-Square Goodness-of-Fit Test.

It is used to determine whether the observed frequencies match an expected distribution.

Ex: Checking whether a die produces six numbers with Equal frequency.

Steps in Chi-Square Test

Step 1: Define the problem.

Clearly state what relationship (or) difference ~~into~~ want to investigate.

Ex: "Is attendance related to examination result?"

Step 2: State the Hypotheses.

H_0 : Attendance and result are Independent.

H_1 : Attendance and result are related.

Step 3: Select Significance level.

usually

$$\alpha = 0.05$$

means a 5% significance level is used.

Step 4: prepare the Observed frequency table.

Ex: Suppose data from 100 students is collected.

Attendance	Pass	Fail	Total
High.	45	5	50
Low	30	20	50
Total	75	25	100.

These are the observed frequencies (O)

Step 5: Calculate Expected frequencies.

The frequency is calculated using:

$$E = \frac{\text{Row Total} \times \text{Column Total}}{\text{Grand Total.}}$$

High Attendance + Pass

$$E = \frac{50 \times 75}{100} = 37.5$$

High Attendance + Fail

$$E = \frac{50 \times 25}{100} = 12.5$$

Low Attendance + Pass

$$E = \frac{50 \times 75}{100} = 37.5$$

Low Attendance + Fail

$$E = \frac{50 \times 25}{100} = 12.5$$

Therefore.

Attendance

Pass

Fail

High

37.5

12.5

low

37.5

12.5

These are the Expected frequencies (E)

Step 6: Calculate Chi-Square Statistic.

formula.

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

Calculate Each cell.

① High + Pass

$$\frac{(45 - 37.5)^2}{37.5} = \frac{7.5^2}{37.5} = \underline{1.5}$$

② High + Fail

$$\frac{(5 - 12.5)^2}{12.5} = \underline{4.5}$$

③ Low + Pass

$$\frac{(30 - 37.5)^2}{37.5} = \underline{1.5}$$

④ Low + Fail

$$\frac{(20 - 12.5)^2}{12.5} = \underline{4.5}$$

Therefore.

$$\chi^2 = 1.5 + 4.5 + 1.5 + 4.5$$

$$\chi^2 = \underline{12}$$

Step 7: Calculate Degrees of freedom.

For a contingency table:

$$df = (r - 1)(c - 1)$$

where,

r = number of rows.

c = number of columns

Here, $r = 2$, $c = 2$.

$\therefore$

$$df = (2 - 1)(2 - 1)$$

$$\boxed{df = 1}$$

Step 8: Find critical value.

Using the chi-square table:

① Significance level = 0.05

② Degree of freedom = 1

By seeing chi-square table.

The critical value is

3.841

Chi-Square table.

df	$\alpha = 0.05$
1	3.841
2	5.991
3	7.815
4	9.488
5	11.070

Step 9: Compare calculated value and critical value.

Calculated value

$$\chi^2 = 12$$

Critical value

$$\chi^2_{\text{critical}} = 3.841$$

Compare

$$12 > 3.841$$

Therefore:

Reject H_0

Note:

Here usually do not say "Accept H_1 ", Instead, we say "Reject H_0 " (or) "Do not reject H_0 ".

because we start with two Hypothesis:

H_0 = Null hypothesis.

H_1 = Alternative hypothesis.

The statistical test is designed to check whether there is enough evidence against H_0 .

So, the decision is based on H_0 .

2.4 Data visualization techniques: histograms, box plots, Scatter plots, heatmaps

Data visualization is the process of representing data in the form of graphs, charts, diagrams and plots.

Large amounts of numerical data can be difficult to understand. Visualization converts the data into a graphical form.

Ex: Suppose the marks of Students
45, 52, 56, 61, 63, 65, 67, 68, 70, 72.

Instead of looking all the numbers, use a graphs to understand how the marks are distributed.

Importance of Data Visualization

1. Easy to understand: Large amounts of data can be understood quickly through graphs.
2. Find patterns: Helps to find patterns and relationships in data.
3. Quick analysis: Helps to analyze large amounts of data.
4. Detects Outliers: Helps in identify unusual values.
5. Better-decision making: provides useful insights for making data-based decisions.
6. Effective communication: Results clearly using graphs and charts.

Major Data Visualization techniques

1. Histogram
2. Box plot
3. Scatter plot
4. Correlation Heatmap

1. Histogram

* A histogram is a graphical representation of the frequency distribution of numerical data.

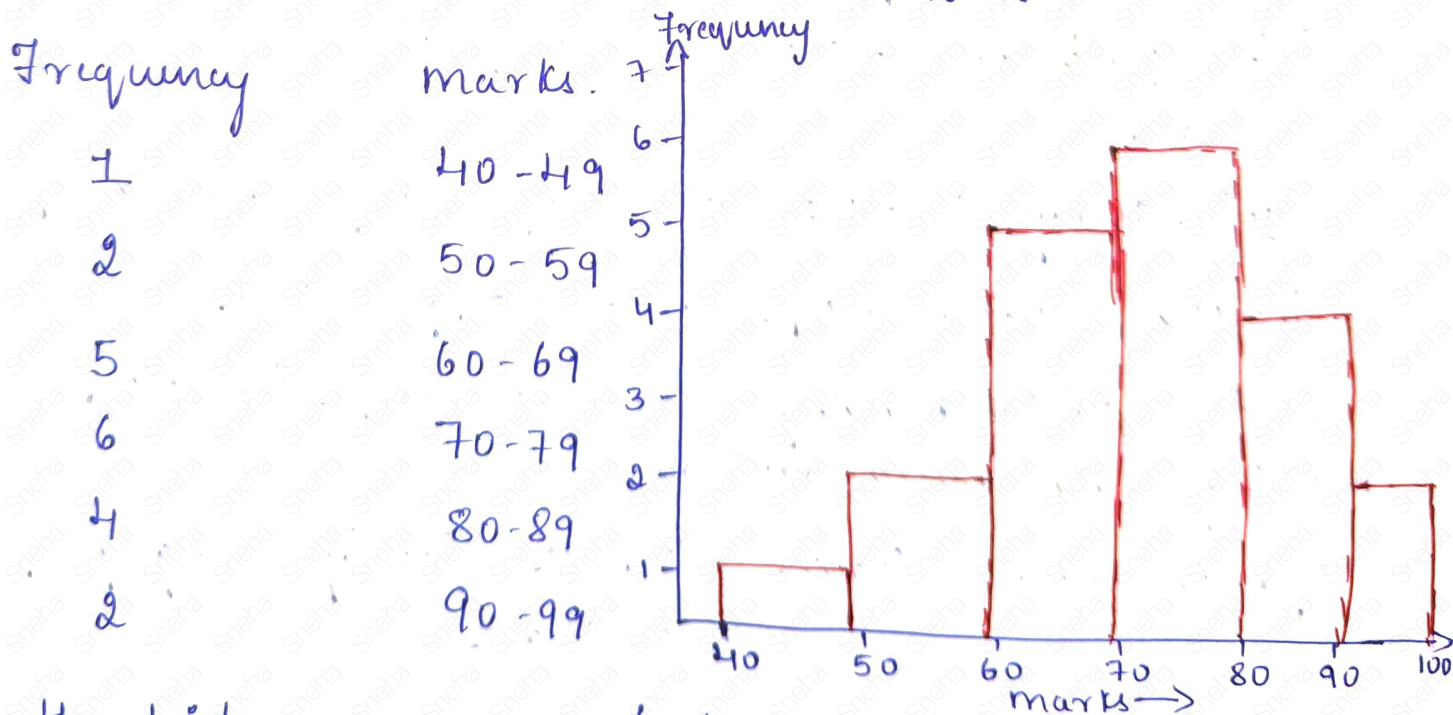
* It divides numerical data into Intervals called bins (or) Class Intervals and show how many observations.

Example:

Consider the marks of 20 Students

45, 52, 56, 61, 63, 65, 67, 68, 70, 72, 73, 74, 76, 78, 80, 82, 85, 87, 90, 92.

Now, divide the marks into the Intervals



So, the histogram represents the number of students in each marks interval.

Code:

```
import matplotlib.pyplot as plt  
marks = [ ]  
  
plt.hist(marks, bins=6)  
plt.xlabel("Marks")  
plt.ylabel("Frequency")  
plt.title("Histogram of Student marks")  
plt.show()
```

2. Box plot

A box plot, also called a box-and-whisker plot, is a visualization technique used to summarize the distribution of numerical data.

It represents the data using,

1. Minimum
2. First Quartile (Q1)
3. Median (Q2)
4. Third Quartile (Q3)
5. Maximum

Example:

Consider the Student data

10, 12, 14, 15, 16, 18, 20, 22, 24, 26, 30

Step 1: Minimum

The smallest value is 10.

Step 2: Median

There are 11 observations

median = 18

Step 3: First Quartile (Q_1)

The lower half is

10, 12, 14, 15, 16.

The middle value is

$$Q_1 = 14$$

Step 4: Third Quartile (Q_3)

The upper half

20, 22, 24, 26, 30

$$Q_3 = 24$$

Step 5: maximum

$$\text{Max} = 30$$

Code:

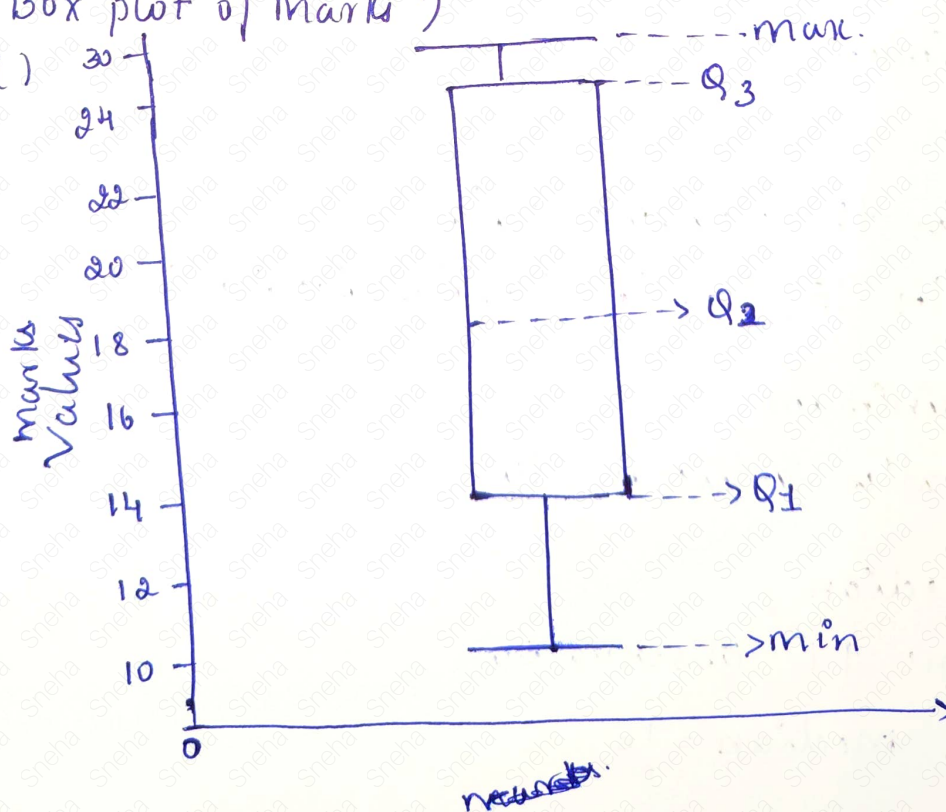
```
import matplotlib.pyplot as plt  
marks = [
```

```
plt.boxplot(marks)
```

```
plt.ylabel("Marks")
```

```
plt.title("Box plot of Marks")
```

```
plt.show()
```



3. Scatter plot

A Scatter plot is a graph used to represent the relationship between two numerical values/variables.

Each observations is represented by a point.

For Example

1. Study hours vs marks
2. Height vs weight
3. Temperature vs Electricity

Taking Study hours vs marks, to see whether students who study more hours tend to get higher marks.

Consider the data.

Student	Study Hours	Marks
A	1	40
B	2	45
C	3	50
D	4	60
E	5	65
F	6	70
G	7	78

Here

Study hours $\rightarrow$ X axis

Marks $\rightarrow$ Y axis

Therefore, Each Student become one point.

$(1, 40), (2, 45), (3, 50), (4, 60), (5, 65), (6, 70), (7, 78)$

Types of relationships

1. positive relationship

points generally move upward, therefore the relation is called positive.

As one variable increases, the other variable tends to increase.

Study Hours $\uparrow$
 $\downarrow$
marks $\uparrow$



2. Negative relationship

points generally move downward, therefore the relation is called negative.

As one variable increases, the other variable tends to decrease.

Price $\uparrow$
 $\downarrow$
Demand $\downarrow$



As price increases, the number of customers decrease.

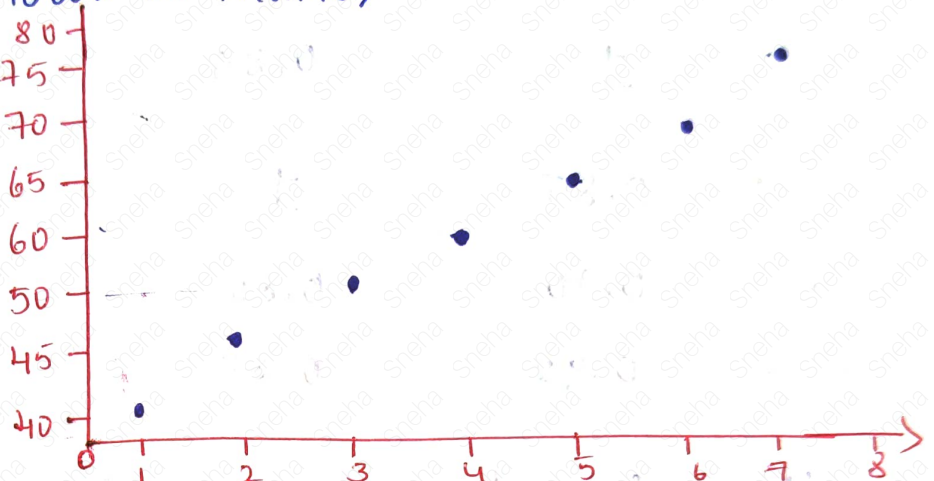
3. No relationship

points are scattered without a clear pattern.



Code:

```
import matplotlib.pyplot as plt  
hours = [1, 2, 3, 4, 5, 6, 7]  
marks = [40, 45, 50, 60, 65, 70, 78]  
plt.scatter(hours, marks)  
plt.xlabel("Study Hours")  
plt.ylabel("Marks")  
plt.title("Study Hours vs Marks")  
plt.show()
```



4. Correlation Heatmap

A heatmap is a graphical representation of numerical data in which different colours (or) shades are used to represent different values.

A correlation heatmap is used to represent the relationship between two variables in a dataset.

The correlation usually ranges from -1 to $+1$.

$+1$ $\rightarrow$ perfect positive correlation

0 $\rightarrow$ No linear correlation

-1 $\rightarrow$ perfect negative correlation

Ex:

0.95 $\rightarrow$ positive

0.10 $\rightarrow$ weak relationship

-0.90 $\rightarrow$ Strong negative

Example:

We have four variables.

Study hours.

Attendance.

Marks.

Exam marks

We get;

	Study hours	Attendance	Assignment marks	Exam mark.
Study hours	1	0.85	0.90	0.92
Attendance	0.85	1	0.80	0.88
Ass. marks	0.90	0.80	1	0.94
Exam mark	0.92	0.88	0.94	1

Here, the diagonal values are always 1 because Each variable is being compared with itself.

Correlation (Study hours, Study hours) = 1.

Every variable has a perfect positive correlation

Interpretation of correlation

+1 → perfect positive correlation

+0.70 to +0.99 → Strong positive.

+0.30 to +0.69 → moderate positive.

0 → weak / no linear

-0.30 to -0.69 → moderate negative.

-0.70 to -0.99 → Strong negative.

-1. → perfect negative correlation.

taking^{above} example.

1. Study hours and attendance, the correlation is 0.90
this indicates a strong positive relationship b/w two variables.

Code.

```
import pandas as pd
import matplotlib.pyplot as plt
```

```
data = {
    'Study Hours': [2, 4, 5, 6, 8],
    'Attendance': [60, 70, 75, 85, 95],
    'Assignment': [65, 70, 78, 85, 92],
    'Marks': [50, 65, 70, 82, 95]
}
```

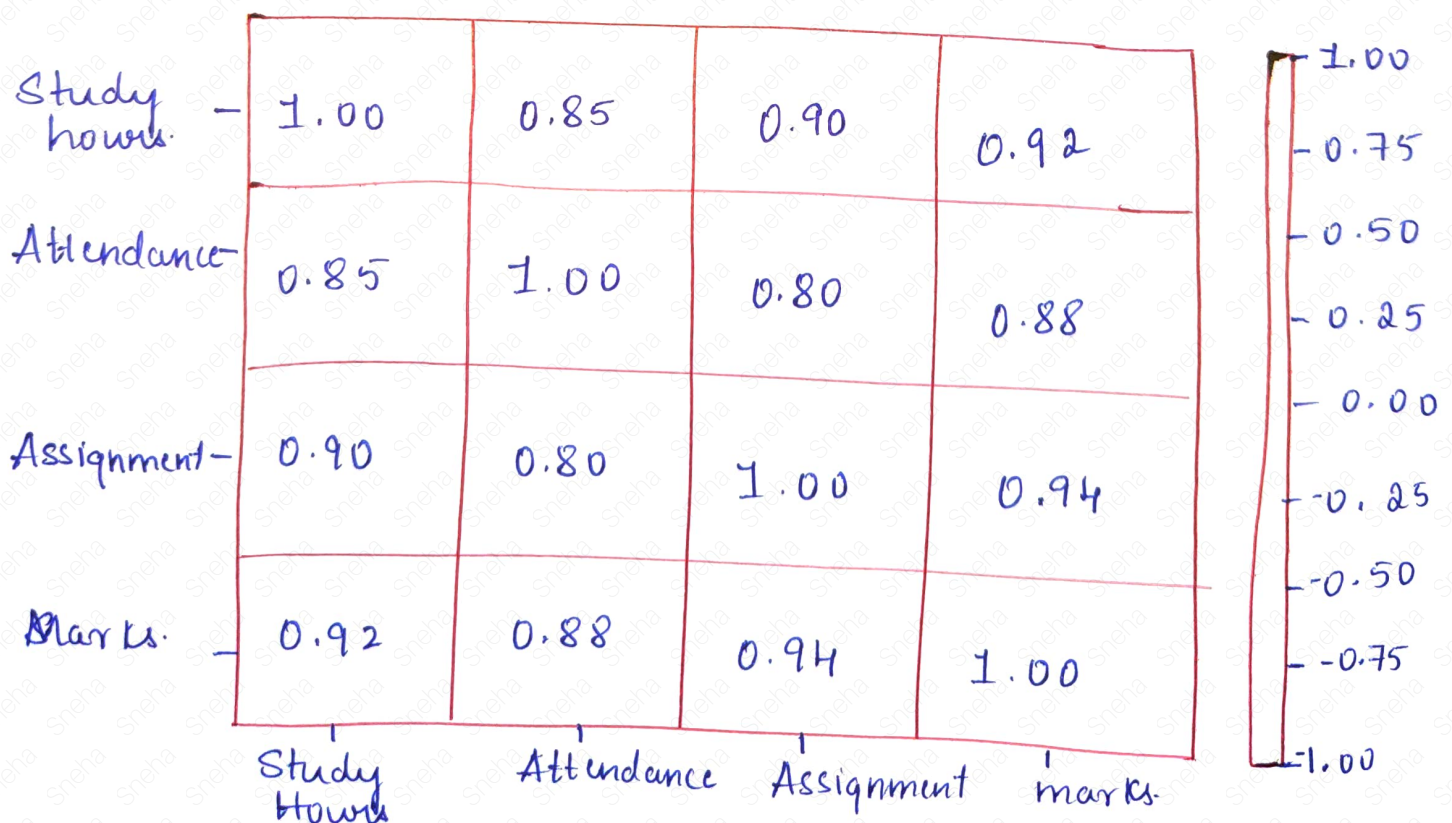
```
df = pd.DataFrame(data)
```

```
Correlation = df.corr()
```

```
Sns.heatmap(Correlation, annot=True, cmap='coolwarm')
```

```
plt.title("Correlation Heatmap")
```

```
plt.show()
```



2.5 EDA using pandas and matplotlib/seaborn

Library

Exploratory Data Analysis is the process of Examining, Summarizing, cleaning and Visualizing a dataset.

EDA help us to understand

1. What type of data is available?
2. How the data is distributed?
3. Are there any missing values?
4. Are there relationships b/w variables? and so on.

Ex: Student data.

Student	Study hours	Attendance	marks
A	2	60	50
B	4	70	65
C	5	75	70
D	6	85	82
E	8	95	95

python libraries used for EDA

Three Important libraries used for EDA are.

1. Pandas.
2. Matplotlib.
3. Seaborn.

1. python Pandas

pandas mainly used to work with tabular data / dataframes.

Import pandas as pd.

It is used for,
load data, inspect data, filter data, Calculate Statistics,
find missing values, remove duplicates.

2. Matplotlib

used for creating different types of graphs and charts.

It can create;

Imported as

import matplotlib.pyplot as plt.

1. line charts
2. Bar charts
3. Histogram
4. Scatter plots
5. Box plots

3. Seaborn

Seaborn is a python visualization library built on matplotlib. It is useful for statistical visualization.

It can create;

import Seaborn as sns

1. pair plots
2. Count plots
3. Heatmaps

1. Import the libraries.

```
Import pandas as pd  
Import matplotlib.pyplot as plt  
Import Seaborn as sns.
```

2. Create the Dataset

```
data = {
```

```
}
```

```
df = pd.DataFrame(data)  
print(df).
```

3. Viewing the dataset

1. head()

The head() function displays the first five rows.

```
df.head()
```

2. tail()

The tail() function displays the last five rows.

```
df.tail()
```

4. Understanding the Structure of data

1. Shape

The Shape attribute gives the number of rows and columns.

```
print(df.shape)
```

Ex: (5, 3).

This means

5 rows

3 columns.

2. Columns

Displays the column name.

```
print(df.columns)
```

3. info()

The info() function provides information about the dataset.

```
print(df.info())
```

It provides:

Number of rows.

Number of columns.

Column names.

4. Data types

```
df.dtypes
```

5. Descriptive Statistics

Used to summarize numerical data.

```
print(df.describe())
```

It provides:

Count → Number of observations

Mean → Average.

Std → Standard deviation

Min → Minimum value.

25% → First quartile (Q1)

50% → Median (Q2)

75% → Third Quartile (Q3)

Max → Maximum value.

6. Checking Missing values

Missing values are data values that are not available.

`df.isnull()`

`df.isnull()` is used to check whether a value is missing or not in a dataset.

`df.isnull().sum()`

`sum()` counts how many missing values are present in dataset.

7. Checking Duplicate Values

Duplicate records are repeated rows in a dataset.

`df.duplicate()`

to count the duplicate record.

`df.duplicate().sum()`

Data Visualization

Data visualization is the graphical representation of data.

Common EDA Visualizations

1. Histogram

2. Box plot

3. Scatter plot

4. Bar plot

5. Correlation heatmap

6. pair plot

} Refer Data visualization techniques.

Complete EDA program

Import pandas as pd

Import matplotlib.pyplot as plt

Import Seaborn as sns.

Create dataset

```
data = {  
    'Study Hours': [2, 4, 5, 6, 8]  
    'Attendance': [60, 70, 75, 85, 95],  
    'Marks': [50, 65, 70, 82, 95]  
}
```

Create Data Frame

```
df = pd.DataFrame(data)
```

```
print(df) # Display dataset
```

First five row

```
print(df.head())
```

Last five row

```
print(df.tail())
```

Shape

```
print(df.shape)
```

Column names

```
print(df.columns)
```

Information

```
print(df.info())
```

Descriptive Statistics

```
print(df.describe())
```

Mean

```
print(df['Marks'].mean())
```

Median

```
print(df['Marks'].median())
```

mode

```
print(df['Marks'].mode())
```

Variance

```
print(df['Marks'].var())
```

Standard deviation

```
print(df['Marks'].std())
```

Missing values.

```
print(df.isnull().sum())
```

Duplicate values.

```
print(df.duplicated().sum())
```

Histogram

```
plt.hist(df['Marks'])
```

```
plt.xlabel('Marks')
```

```
plt.ylabel('Frequency')
```

```
plt.title('Distribution of marks')
```

```
plt.show()
```

Box plot

```
Sns.boxplot(y=df['Marks'])
```

```
plt.title('Box plot of marks')
```

```
plt.show()
```

Scatter plot

```
Sns.scatterplot(study hours, marks)
```

```
plt.xlabel('Study Hours')
```

```
plt.ylabel('Marks')
```

```
plt.title('Study Hours Vs Marks')
```

```
plt.show()
```

Correlation matrix

```
print(df.corr())
```

Heatmap

```
Sns.heatmap(df.corr(), annot=True)
```

```
plt.title('Correlation Heatmap')
```

```
plt.show()
```

Pair plot

```
Sns.pairplot(df)
```

```
plt.show()
```

Q.6 Univariate, bivariate and multivariate analysis.

Data Analysis is the process of examining data to find useful information, patterns, ~~and~~ relationships and trends.

Based on the number of variables considered at a time, data Analysis can be classified into three types.

1. Univariate Analysis [Analysis of one variable]
2. Bivariate Analysis [Analysis of two variables]
3. Multivariate Analysis [Analysis of three or more variables]

Example:

Consider the student dataset.

Student	Study Hours	Attendance	Assignment
A	2	60	55
B	4	70	65
C	5	75	70
D	6	85	80
E	8	95	90

1. Univariate Analysis

Univariate analysis is the analysis of a single variable at a time.

The main purpose is to understand the distribution, central tendency, spread, and characteristics of one variable.

Example:

We want to analyze only the marks of students.

Marks: 50, 65, 70, 82, 95

Here, considering only one variable - marks.

Measures used in Univariate Analysis.

Common measures include.

- | | | | |
|-----------|------------|-------------|-----------------------|
| 1. Mean | 3. mode | 5. maximum | 7. Standard deviation |
| 2. Median | 4. minimum | 6. Variance | 8. percentiles |

Ex: mean

$$\text{mean} = \frac{50 + 65 + 70 + 82 + 95}{5} = \frac{362}{5} = 72.4$$

Visualization in Univariate analysis

- | | |
|--------------|--|
| 1. Histogram | } <u>Rel or</u>
<u>Data visualization (2.4)</u> |
| 2. Box plot | |
| 3. Bar chart | |

2. Bivariate Analysis.

Bivariate Analysis is the analysis of two variables at the same time.

The main purpose is to find whether there is a relationship (or) dependency between two variables.

Example:-

Relationship between study hours and marks.

Study Hours

Marks

2

50

4

65

5

70

6

82

8

95

Here, we are considering two variables.

- Study Hours
- Marks

Visualization in Bivariate Analysis.

1. Scatter plot
2. Correlation

} Ref (2.4)

3. Multivariate Analysis.

Multivariate analysis is the analysis of three (or) more variables simultaneously.

It is used to understand relationship and patterns involving multiple variables.

Ex: Student data like.

1. Study Hours
2. Attendance
3. Assignment Marks
4. Final Marks

Here, more than two variables are considered, so it is multivariate analysis.

Visualization in Multivariate Analysis.

1. pair plot
2. Correlation Heatmap

} Ref (2.4)